NumPy and Pandas Basics

Python course - Lesson 2+3

TALIA MORCHI

ACADEMY

Previous Lesson Topics

- Python language – brief introduction
- Data types Strings – manipulating strings and regular expressions.
- Basic flow control - conditions and loops
- Data structures - lists, dictionaries, sets and tuples

Today's Topics

- NumPy
  - NumPy array
  - NumPy matrix
- Pandas
  - Pandas Series
  - Pandas DataFrame

NumPy

TALIA MORCHI

ACADEMY

Introduction to NumPy

Numerical Python

The library is installed as part of the Anaconda installation

An open source library used in almost all fields of science and engineering

Provides quick mathematical calculations on arrays and matrices

Import Library

In order to use a library, we will have to import it into the notebook

Library name

```
import numpy as np
```

Library nickname

1D NumPy Array

- NumPy array is 1 dimensional list
    - NumPy array can hold only one data type
        - The advantage in NumPy array is that each calculation will be performed on all items in the array
            - Retrieving value from array item in specific index: array_name[index]

1D NumPy Array

- Creating NumPy array:

  arr_name = np.array ([val_1, val_2, … val_n])

- Retriving data:

  arr_name[index]

arr_name	val_1	val_2	val_3	val_4	val_5	val_6
	arr_name[0]	arr_name[1]	. . .	. . .	. . .	arr_name[-1]

1D NumPy Array Slicing

With slicing we can get part of the array items

arr_name[start : stop : step]

Beginning of the slicing

End position (not include)

Step allow skipping on items

1D NumPy array

Condition on array

- Boolean results for all items
- Items for which the condition is true

2D NumPy Array - Matrix

- 2 dimensional NumPy array is called Matrix
  - NumPy matrix can hold only one data type
    - The advantage in NumPy matrix is that each calculation will be performed on all items in the matrix
      - Retrieving value from array item in specific index: matrix _name[row][column]

2D NumPy Array - Matrix

- Creating NumPy array:

```
mtx_name = np.matrix ( [   [val_1_1, val_1_2…, … val_1_n],
                           [val_2_1, val_2_2…, … val_2_n],
                                    . . .
                           [val_n_1, val_n_2…, … val_n_n]   ] )
```

mtx_name

val_1_1	val_1_2	val_1_3	val_1_4	. . .	val_1_n	[0]
val_2_1	val_2_2	val_2_3	val_2_4	. . .	val_2_n	[1]
val_3_1	val_3_2	val_3_3	val_3_4	. . .	val_3_n	[-1]
[0]	[1]	. . .	. . .	. . .	[-1]	

2D NumPy Array - Matrix

- Retriving data:

  mtx_name[row][column]

mtx_name

val_1_1	val_1_2	val_1_3	val_1_4	. . .	val_1_n	[0]
val_2_1	val_2_2	val_2_3	val_2_4	. . .	val_2_n	[1]
val_3_1	val_3_2	val_3_3	val_3_4	. . .	val_3_n	[-1]
[0]	[1]	. . .	. . .	. . .	[-1]	

Array / Matrix Attributes

Attribute	Action	Example
ndim	Number of dimensions	a.ndim, m.ndim
shape	(number of rows, number of columns)	a.shape, m.shape
size	How many total items	a.size, m.size
dtype	Data type of the items	a.dtype, m.dtype

* Attributes does not require parenthesis ()

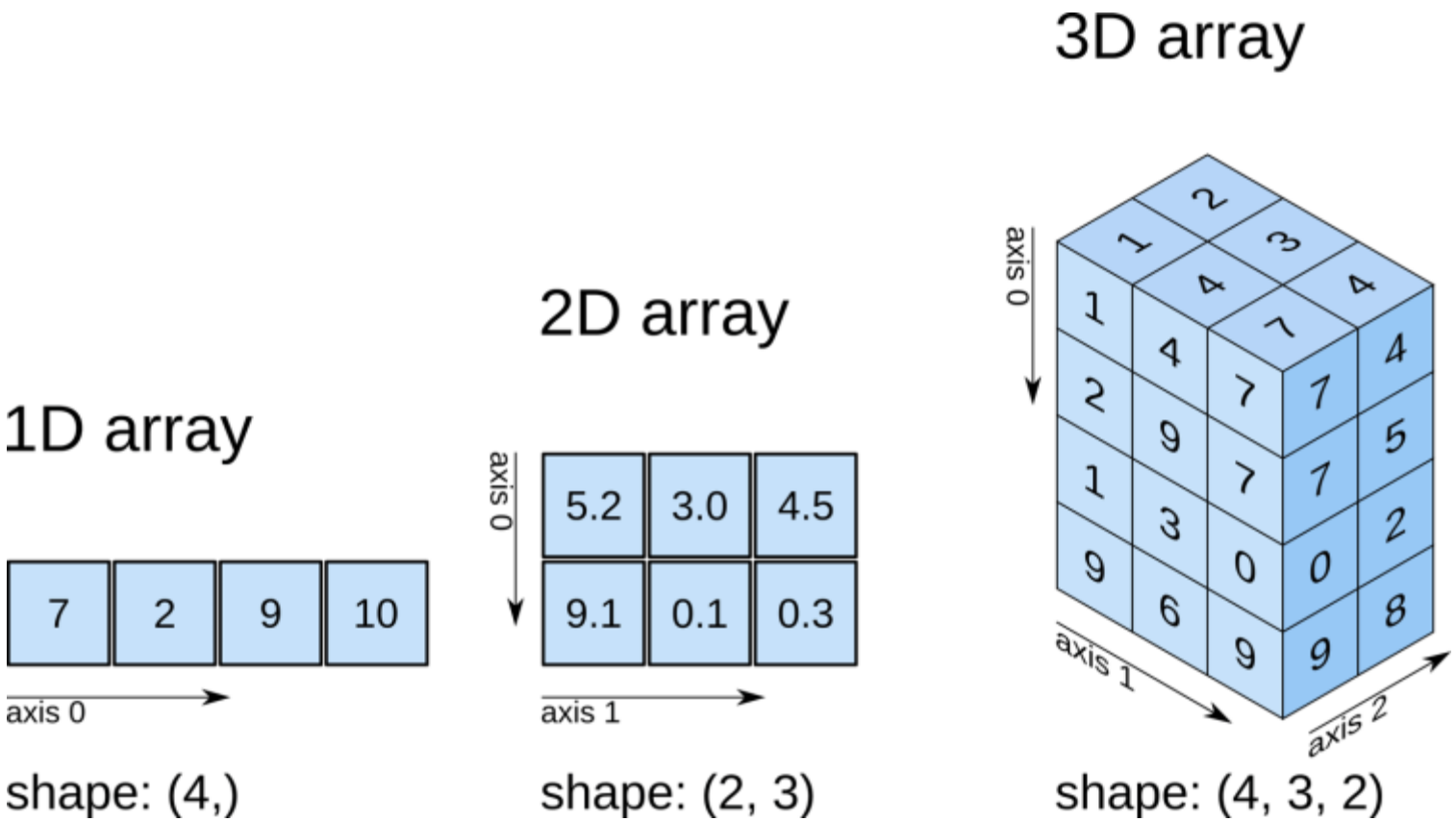
NumPy Methods

Method	Action
np.arange (start, stop, step)	Return evenly spaced values within a given interval
np.linspace (start, stop, no_items)	Return evenly spaced numbers over a specified interval
np.random.rand (d0, d1, ... dn)	Return a sample/s from the uniform distribution over [0, 1)
np.random.randn (d0, d1, ... dn)	Return a sample/s from the standard normal distribution
np.random.randint (low, high, no_items)	Return random integers from low (inclusive) to high (exclusive)

Array / Matrix Methods

Method	Action
max ()	Return the maximum along a given axis
argmax ()	Return indices of the maximum values along the given axis
min ()	Return the minimum along a given axis
argmin ()	Return indices of the minimum values along the given axis
sort ()	Sort an array in-place
sum()	Return the sum of the array elements over the given axis

Array / Matrix



Pandas

TALIA MORCHI

ACADEMY

Pandas

An open source library used primarily for data analytics and data science

Built on NumPy

The library is installed as part of the Anaconda installation

Serves as an excellent tool for analyzing, handling, filtering and sorting data

Import Library

In order to use a library, we will have to import it into the workbook

Library name

```
import pandas as pd
```

Library nickname

Pandas Series

Represents a single column of data

Index

Series name

	A
0	A0
1	A1
2	A2
3	A3
4	A4

Series values

* Index can be changed to any value (string, integer...) and do not have to be unique

Pandas DataFrame

Represents a matrix of data

DataFrame

Index

Series name

	A	B	C
0	A0	B0	C0
1	A1	B1	C1
2	A2	B2	C2
3	A3	B3	C3
4	A4	B4	C4

Series values

* Index can be changed to any value (string, integer...) and do not have to be unique

Import Data From CSV File

pd.read_csv ('file_name.csv')

- Read a comma-separated values (csv) file into DataFrame

- Notebook and file in the same folder:

```
pd.read_csv ('file_name.csv')
```

- Notebook and file are not in the same folder:

```
pd.read_csv (r'full_path\file_name.csv')
```

read_csv () Usefull Parameters

pd.read_csv ('file_name.csv')

Parameter	Action
usecols = [list_with_columns_to_import]	Import the listed column only
squeeze = true	If there is only one column, import data as a Series

Attributes vs Method vs Build in Functions

Attribute – information on the object

object.attribute_name

Method – operation we perform on the object

object.method_name ()

Function – piece of code that we can send object as parameter

function_name (object)

Series / DataFrame Attributes

Attribute	Action	s/df *
values	Returns array with the values	s, df
index	Returns description of the indexes	s, df
dtypes	Returns the dtypes	s, df
is_unique	Return boolean if values in the object are unique	s
name	Returns the Series name	s
shape	Return a tuple representing the dimensionality of the DataFrame	s, df

TALIA MORCHI

* s = can be applied on Series, df= can be applied on DataFrame

ACADEMY

Series / DataFrame Methods

Method	Action	s/df *
head (n)	Return the first `n` rows	s, df
tail (n)	Return the last `n` rows	s, df
max ()	Return the maximum of the values for the requested axis	s, df
argmax ()	Return the row number of the maximum value	s
idxmax ()	Return the index label of the maximum value	s
min ()	Return the minimum of the values for the requested axis	s, df
argmin ()	Return the row number of the minimum value	s, df
idxmin ()	Return the index lable of the minimum value	s
describe ()	Generate descriptive statistics	s, df
sort_values ()	Sort by the values	s, df
sort_index ()	Sort Series by index labels	s, df

TALIA MORCHI

* s = can be applied on Series, df= can be applied on DataFrame

ACADEMY

Series / DataFrame Methods

Method	Action	s/df *
sum ()	Return the sum of the values for the requested axis	s, df
count ()	Count non-NA cells for each column or row.	s, df
mean ()	Return the mean of the values for the requested axis	s
std ()	Return sample standard deviation over requested axis	s, df
value_counts()	Return a Series containing counts of unique values	s
iloc [start : stop : step]	Return the row data in the location	s, df
loc [start : stop : step]	Returns the rows that their *label index* is in the range	s, df
info ()	Print a concise summary of a DataFrame	s, df
keys ()	Return all Series names	s, df
fillna (str/value)	Fill NA/NaN values using the specified method	s, df

* s = can be applied on Series, df= can be applied on DataFrame

Series / DataFrame Methods

Method	Action	s/df *
drop ()	Drop specified labels from rows or columns	s, df
astype ()	Cast a pandas object to a specified dtype ``dtype``	s, df
stack ()	Stack the prescribed level(s) from columns to index	df
unstack ()	Stack the prescribed level(s) from columns to index	df
dropna ()	Remove rows / columns with missing values	s, df

TALIA MORCHI

* s = can be applied on Series, df= can be applied on DataFrame

ACADEMY

Function on Series / DataFrame

Function	Action	s/df *
dir (s)	Return a list of all built-in function and method	s, df
len (s)	Return the number of items in a container	s, df
pd.DataFrame ()	Receive matrix object and return DataFrame object	s

* s = can be applied on Series, df= can be applied on DataFrame

s[] vs iloc [] vs loc[]

s[lable/location] = returns the value with the lable or in that location

s.iloc[location] = returns the value in the location

s.loc[lable] = returns the row/s with thin index lable

Label Index

Location

		A
0	0	A0
1	2	A1
2	6	A2
3	10	A3
4	20	A4

s[6] = s.iloc[2] = s.loc[6] = 'A2'

Concatenation

pd.concat ([df_1,df_2,…df_n])

- Concatenate pandas objects along a particular axis with optional set logic along the other axes.

Parameter	Action
axis = 1	The axis to concatenate along: 0 = index, 1 = columns. default 0
keys=['x', 'y']	concatenate df use multi index x = multi index for df_1, y=multi index for df_2
ignore_index	Creating new index for the new Dataframe

Appending

df_1.append(df_2)

- The append() function is used to append rows of other to the end of caller, returning a new object.
- Columns in other that are not in the caller are added as new columns

Parameter	Action
ignore_index	Creating new index for the new Dataframe

Merging

df_1.merge(df_2, how = 'inner', on = ‘column_name')

- Merge DataFrame or named Series objects with a database-style join
- The join is done on columns or indexes.

Merging

df_1.merge(df_2, how = 'inner', on = ‘column_name')

Parameter	Action
how = 'outer'	Optional values: inner, left, right, merge
on = [‘Col_x’, ‘Col_x’]	Column or index level names to join on
suffixes = [‘ – A’, ‘ – B’]	Suffix to apply to overlapping column names in the left and right side
ignore_index	Creating new index for the new Dataframe
indicator = True	If True, adds a column to output DataFrame called "_merge" with information on the source of each row.

Thank You

TALIA MORCHI

morchi.talia@gmail.com

ACADEMY